

DIFFERENCES IN EXERCISE CAPACITY AND METABOLIC HEALTH BETWEEN ADULTS WITH AND WITHOUT TYPE 2 DIABETES

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INTRODUCTION

Type 2 diabetes (T2D) is associated with reduced exercise capacity, obesity, and poorer glycemic control, all of which may contribute to greater cardiometabolic risk and reduced physical function [1, 2]. Characterising these differences is important for informing targeted exercise prescription and risk-reduction strategies. This study compared exercise capacity, body composition, and metabolic health between adults with and without T2D.

METHODS

Adults with T2D (T2D; n = 21) and adults without T2D (controls; n = 21) participated in this cross-sectional study. Participant characteristics included age, sex, height, weight, body mass index (BMI), systolic and diastolic blood pressure (SBP and DBP), and resting heart rate (RHR). Body composition was assessed using dual-energy X-ray absorptiometry (DXA), while exercise capacity was evaluated using maximal cardiopulmonary exercise testing (CPET; controls, n = 20; T2D, n = 21), the 6-minute walk test (6MWT; controls, n = 21; T2D, n = 20), and the 2-minute step test (2MST; controls, n = 21; T2D, n = 19). Fasting blood samples were analysed for glycaemic control (HbA1c), lipid profile (total cholesterol, LDL-C, HDL-C, and triglycerides), and fasting insulin. Between-group differences were assessed using independent-samples t-tests or Mann–Whitney U tests, as appropriate. General linear models were used to compare outcomes between groups after adjustment for age, sex, and BMI. Statistical significance was set at $p < 0.05$.

RESULTS AND DISCUSSION

Forty-two adults were included (T2D, n = 21; controls, n = 21). Compared with controls, adults with T2D were older (62.57 ± 8.20 vs 50.90 ± 11.96 years) and had higher BMI [31.08 ($24.21, 34.46$) vs 23.56 ($21.98, 26.93$) kg/m^2], total body fat, body fat percentage, SBP, RHR, HbA1c, fasting insulin, and triglycerides, but lower total cholesterol, HDL-C, and LDL-C (all $p < 0.05$). Despite this less favourable adiposity profile, adults with T2D had numerically higher total bone mineral density and

muscle mass than controls, although these differences were not statistically significant ($p = 0.23$ and $p = 0.33$, respectively). In unadjusted analyses, adults with T2D demonstrated significantly lower absolute and relative peak oxygen uptake ($\text{VO}_{2\text{peak}}$), lower maximal heart rate (HR_{max}), shorter CPET time to exhaustion, and poorer performance on 6MWT and 2MST. After adjustment for age, sex, and BMI, adults with T2D continued to demonstrate significantly lower absolute $\text{VO}_{2\text{peak}}$ compared with controls (2853.68 vs 2129.34 $\text{mL}\cdot\text{min}^{-1}$; adjusted mean difference, 724.35 $\text{mL}\cdot\text{min}^{-1}$; 95% CI, $490.52 - 958.17$; $p < 0.001$) (Table 1). Relative $\text{VO}_{2\text{peak}}$ also remained significantly lower in adults with T2D (38.26 vs 29.49 $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$; adjusted difference 8.78 $\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$; 95% CI, $5.15 - 12.41$; $p < 0.001$). However, after adjustment for the selected covariates, no significant between-group differences were observed in CPET time to exhaustion ($p = 0.09$), 6MWT distance ($p = 0.08$), or 2MST performance ($p = 0.66$).

CONCLUSIONS

Adults with T2D exhibited poorer exercise capacity, body composition, and metabolic health than adults without T2D. These findings reinforce the importance of lifestyle interventions targeting cardiorespiratory fitness and body composition to improve cardiometabolic health and physical function in adults with T2D.

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REFERENCES

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Table 1: Adjusted between-group differences in cardiopulmonary exercise test outcomes

Variable	Control adjusted mean (95% CI)	T2D adjusted mean (95% CI)	Adjusted difference (95% CI)	p-value
Absolute $\text{VO}_{2\text{peak}}$ ($\text{mL}\cdot\text{min}^{-1}$)	2853.68 (2696.89 – 3010.47)	2129.34 (1972.09 – 2286.58)	724.35 (490.52 – 958.17)	<0.001
Relative $\text{VO}_{2\text{peak}}$ ($\text{mL}\cdot\text{kg}^{-1}\cdot\text{min}^{-1}$)	38.26 (35.83 – 40.70)	29.49 (27.05 – 31.93)	8.78 (5.15 – 12.41)	<0.001
CPET time to exhaustion (s)	953.38 (895.49 – 1011.27)	877.56 (814.60 – 940.52)	75.82 (-13.62 – 165.26)	0.094